

CHOOSECOLS Function in Excel

1. What is CHOOSECOLS?

The CHOOSECOLS function returns only the columns you pick from a table, in the order you choose. It solves the problem of wide tables that have more columns than you need. A Data Analyst uses it to build compact tables, reorder columns, and pull columns that are not next to each other, without copying and pasting.

2. Syntax

`=CHOOSECOLS(array, col_num1, [col_num2], ...)`

Argument	Required?	What it accepts
array	Required	The range or table to take columns from (for example A1:D5)
col_num1	Required	The number of the first column to return. A positive number counts from the left, and a negative number counts from the right (-1 is the last column).
col_num2, ...	Optional	More column numbers. They can be in any order, and the same number can be used more than once.

Column numbers count inside the array, not by the sheet column letters. In B1:E5, column number 1 is column B. If you include the heading row in array, the headings come along with the data.

3. Simple Example

Return only the Product and Sales columns:

	A	B	C	D	E	F
1	Product	Category	Sales		Product	Sales
2	Laptop	Electronics	1200		Laptop	1200
3	Mouse	Electronics	650		Mouse	650
4	Chair	Furniture	300		Chair	300
5	Desk	Furniture	750		Desk	750

Formula in E1 only: `=CHOOSECOLS(A1:C5, 1, 3)`

Excel takes column 1 (Product) and column 3 (Sales) from A1:C5 and skips column 2 (Category). The headings in row 1 come along because row 1 is in the array. **Result: 2 columns and 5 rows**

4. Another Example

Return Salary first, then Name, counting Salary from the right:

	A	B	C	D	E	F	G
1	ID	Name	Dept	Salary		Salary	Name
2	E101	Ayesha	IT	50000		50000	Ayesha
3	E102	Rahim	HR	42000		42000	Rahim
4	E103	Nabila	IT	61000		61000	Nabila
5	E104	Tanvir	Sales	38000		38000	Tanvir

Formula in F1 only: `=CHOOSECOLS(A1:D5, -1, 2)`

The -1 means the last column of the array (Salary). The 2 means the second column (Name). Excel returns the columns in the order you list them, so Salary comes before Name.

5. When to Use It

- Pull only the columns you need from a wide dataset for a report or dashboard
- Reorder the columns of a table without moving the source data
- Return columns that are not next to each other, such as the first and fourth
- Always return the last column of a table, even when columns are added (-1)
- Select columns from the result of another function, such as FILTER or SORT

6. Important Things to Remember

- CHOOSECOLS is a dynamic array function. One formula fills many cells, so the cells for the result must be empty, or Excel shows #SPILL!.
- Positive column numbers count from the left. Negative column numbers count from the right.
- Columns appear in the order you list them. Repeating a column number repeats that column.
- A column number of 0, or a number larger than the number of columns, gives #VALUE!.
- CHOOSECOLS returns values only, not the source formatting. Format the result separately.
- CHOOSECOLS is available in Microsoft 365 and Excel 2024. In Excel 2021 or earlier it shows #NAME?.

7. Common Mistakes

Mistake: Using the sheet column number instead of the position inside the array, such as 4 for column D in B1:D5, which gives #VALUE!. **Fix:** Count from the first column of the array. In B1:D5, column D is number 3.

Mistake: Using 0 as a column number. **Fix:** Use 1 or more to count from the left, or -1 or less to count from the right.

Mistake: Getting #SPILL! because cells where the result will go already contain data. **Fix:** Clear those cells or move the formula to an empty area.

Mistake: Expecting the number formats and colors of the source to carry over. **Fix:** Apply the formatting to the result range yourself.

8. Data Analyst Tip

Combine CHOOSECOLS with other functions. For example, `=CHOOSECOLS(FILTER(A2:D5, C2:C5="IT"), 2, 4)` returns only the Name and Salary of the IT employees: Ayesha 50000 and Nabila 61000. This gives a compact report in one formula.